# Efficient Two-Way Edge Backscatter with Commodity Bluetooth

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***Abstract*—Two-way backscatter is essential to general-purpose backscatter communication as it provides rich interaction to support diverse applications on commercial devices. However, existing Bluetooth backscatter systems suffer from unstable uplinks due to poor carrier-identification capability and inefficient downlinks caused by packet-length modulation. This paper proposes EffBlue, an efficient two-way backscatter design for commercial Bluetooth devices. EffBlue employs a simple edge backscatter server that alleviates the computational burden on the tag and helps build efficient uplinks and downlinks. Specifically, efficient uplinks are designed by introducing an accurate synchronization scheme, which can effectively eliminate the use of non-compliant packets as carriers. To break the limitation of packet-level modulation, we design a new symbol-level WiFi-ASK downlink where the edge sends ASK-like WiFi signals and the tag can decode such signals using a simple envelope detector. We prototype the edge server using commodity WiFi and Bluetooth chips and build two-way backscatter tags with FPGAs. Experimental results show that EffBlue can identify the target excitations with more than 99% precision. Meanwhile, its WiFi-ASK downlink can achieve up to 124 kbps, which is 25x better than FreeRider.**

## I. Introduction

In recent years, the rapid development of the Internet of Things (IoT) has led to the widespread deployment of sensors in our daily life. Backscatter technology [1]–[11] offers a solution to this problem due to its ultra-low-power consumption. By utilizing a battery-free design, sensor nodes can harvest energy from the environment and transmit data using ambient excitations. However, mature RFID backscatter systems have been hindered by the high costs of readers. Therefore, researchers have been exploring various backscatter paradigms based on commodity radios(e.g., WiFi [12]–[15], BLE [16], [17], Zigbee [17], [18], etc.), aiming to turn backscatter into general-purpose communication technique.

In order to achieve compatibility with commodity radios like Bluetooth and WiFi, it is necessary for the tag to identify the target ambient carriers for an efficient uplink. Additionally, a downlink is crucial for coordinating tags, making two-way communication essential. The state-of-the-art work [1], [6]–[8], [12]–[22] have made a great step in pushing the current backscatter system towards a general-purpose system. HitchHike [12] and FreeRider [17] propose codeword translation technique with commodity carriers in the uplink. To transmit coordination messages to tag, they introduce packet length modulation in the downlink. Interscatter [18] and RBLE [21] adopt dewhitened BLE packets as carriers, but their downlink lacks a mechanism for target carrier identification.

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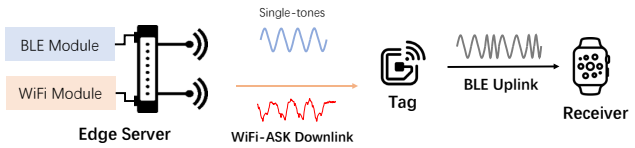


Fig. 1. EffBlue design. EffBlue employs an edge to help tag achieve two-way communication: By adopting a single-tone identification scheme, the tag achieves an efficient BLE uplink. Meanwhile, the tag can also receive data efficiently with the WiFi-ASK downlink.

Existing systems have made some progress in exploring two-way backscatter communication. However, current backscatter systems still suffer from the following limitations that hinder further advancements: *1) Unreliable uplink*. Backscatter systems have their appetite for excitations among ambient packets. For example, RBLE modulates on dewhitened packets (single-tone) to regenerate BLE signals. However, its modulation process is triggered by energy level, which means if non-single-tone packets come, the modulation is performed too, thus cutting down the uplink efficiency. *2) Inefficient downlink.* To enable valid modulation, tag needs to know the carrier frequency. FreeRider and RBLE adopt packet-length modulation (PLM) scheme, which encodes bit '0' and '1' with different packet lengths, and the application is limited. For example, the commodity BLE carrier can change the frequency for each packet. It is impossible to forward the carrier information through a packet-level downlink. Consequently, when dealing with high-throughput radios, tags suffer from low efficiency due to these limitations.

To address the above issues, we propose EffBlue, an efficient two-way backscatter system for commercial Bluetooth devices as shown in Fig. 1. EffBlue improves the modulation efficiency by identifying the target carrier in the uplink. Further, it proposes an efficient downlink to configure the tag dynamically. This two-way communication enables the tag to utilize the diverse BLE signals, boosting the transmission performance. To cut down the computational burden of tag, EffBlue adopts an edge server to help build this two-way communication. The key design of EffBlue is as follows.

- **Reliable identification-based uplink.** To make transmis-


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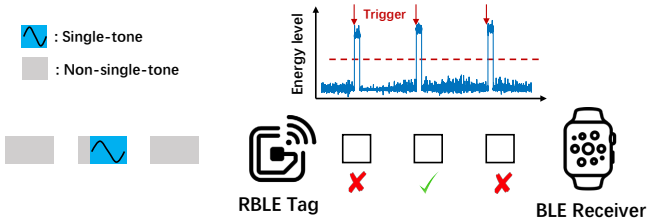


Fig. 2. RBLE tag could be triggered to modulate by arbitrary excitation packets. Thus receiver fails to receive the backscattered signal when the tag modulates on a non-single-tone.

sion valid, tag has to modulate data on the target carrier. This is especially important when dealing with commodity carriers as they inevitably involve non-target carrier packets. EffBlue utilizes periodic advertisements as the carrier, which has better carrier availability. To address the issue of false excitations due to energy-based identification, we propose a synchronization scheme based on carrier characteristics. we use carrier packet lengths to identify target excitations. After identification, we predict the next excitations based on periodic intervals. Unfortunately, those intervals could drift far from the predefined periodicity due to hardware imperfection. We further introduce iterative synchronization based on the prediction window and improve the accuracy of single-tone identification in the tag's uplink modulation.

- **Efficient WiFi-ASK downlink.** The commodity BLE carrier packets are transmitted among different channels. Therefore, the tag needs to know the carrier frequency. The downlink should support conveying the carrier information during packet intervals. To break the limitation of packet-level downlink, we design a new symbol-level WiFi-ASK downlink. The edge encodes data in ASK-like WiFi signals. Tag adopts an envelope detector to extract the signal. After getting the signal envelope, tag applies averaging to remove the high-frequency signal. Then tag quantizes it using an adaptive threshold and decodes out data of each efficient WiFi-ASK downlink symbol. Thus we enable tag to decode data efficiently with this high-throughput downlink.

We prototype EffBlue using commercial radios, and customized tags. Through extensive empirical evaluation, our main results are summarized as follows.

1) When the ratio of non-single tones to single tones is 1:1 in excitation signals, EffBlue achieves more than 99% precision in identifying periodic single tones, while RBLE only achieves 51.9%.
2) With EffBlue's accurate synchronization scheme, 96% synchronization errors are within 2 $\mu s$.
3) Compared to PLM, EffBlue achieves up to 124 kbps, which is 25x better than FreeRider.
4) With a redundant encoding scheme, the WiFi-ASK downlink BER can be reduced from 16.52% using S = 1 encoding to 0.43% using S = 8 encodings.

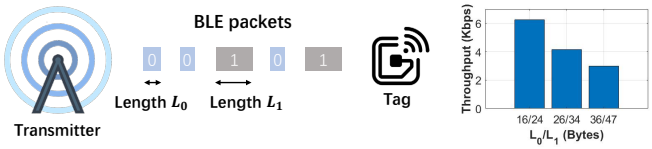


Fig. 3. PLM-based downlink encodes bit "0" and "1" with length $L_0$ and $L_1$ respectively.

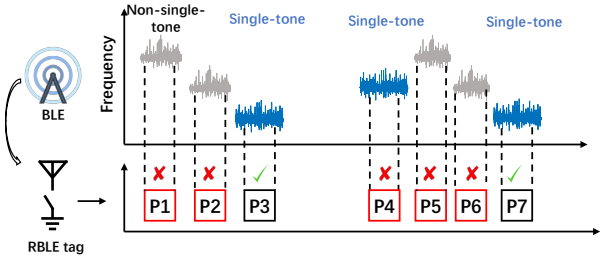


Fig. 4. RBLE modulates on all coming packets, considering them as carriers from a single channel. Only P3, P7 that are backscattered on single tones from the expected channel are valid.

## II. Motivation

FreeRider [17] and RBLE [21]'s modulation processes are triggered by the energy level and they use a PLM based downlink. Two limitations remain in such Bluetooth backscatter systems. One is the false-positive cases of detection during uplink transmission and the other is the inefficient downlink. They motivate the work in this paper.

1) **Unreliable uplink.** RBLE utilizes BLE packets as excitation signals. It dewhitens the excitation payload to constant 0s or 1s in PHY layer to produce single tones and modulates them to regenerate new BLE packets. Utilizing single tones as excitation signals improves the communication link quality. But it requires the tag to modulate the correct position of the excitation signal. RBLE takes an envelope detector to monitor the energy level. When packets arrive, the energy level will exceed the predefined threshold, and tag starts modulation. This mechanism leads to the issue that arbitrary packets could trigger tag's modulation process. As shown in Fig. 2, there are single tones and non-single tones in the ambient environment. Backscattered signals on the non-single tones are invalid and commodity BLE devices cannot decode them. Fig. 7 shows that when non-single tones/single tones are 1:1 in the excitation signals, half of the modulations are invalid. These false-positive errors greatly reduce the modulation efficiency of tag.
2) **Inefficient downlink.** For low power consumption tag, decoding Frequency-shift keying (FSK) signals would be time and resource-consuming. However, the Amplitude-shift keying (ASK) signal would be much easier, since tag can utilize an envelope detector to extract the amplitude information. Generating an ASK signal is simple using SDR devices, but has a high cost. Thus, state-of-the-art work adopts commodity devices as transmitters. For


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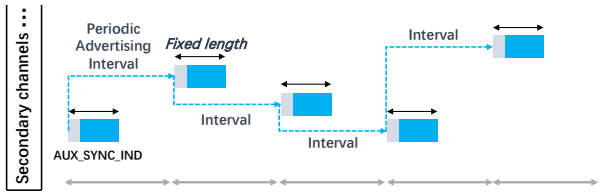


Fig. 5. The payload of periodic advertisement signal AUX_SYNC_IND is controlled to single-tone. And they are observed to have fixed length and interval.

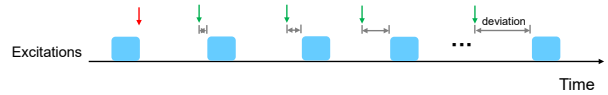


Fig. 6. Time drift of prediction. Red arrows denote the recognized packet with expected length. The green arrow denotes the predicted excitation packet.

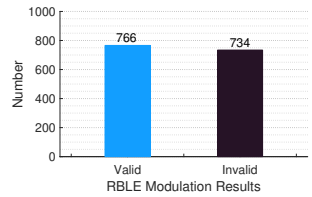


Fig. 7. False excitations reduce uplink efficiency.

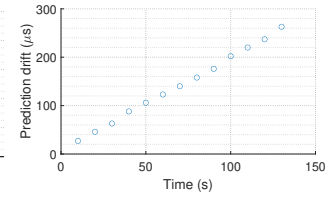


Fig. 8. Prediction drift accumulates over time.

example, FreeRider and RBLE take the packet-length modulation (PLM) scheme in the downlink. PLM has good performance when encountering interference as the packet length does not degrade with distance. But it suffers from low transmission efficiency. We use common BLE advertisement packets to encode data as shown in Fig. 3. The smallest empty packet has a duration of 10 bytes (80 $\mu s$, including a preamble and CRC). The longest packet (not extended length because generating packets of extended length is more complex) has a duration of 47 bytes (376 $\mu s$). The average time of a PLM symbol is $(80 + 376)/2 = 228$ $\mu s$. Fig. 3 shows the theoretical throughput of different combinations of packet length. Results show that PLM remains in a low throughput in different cases.

Having a limited power budget, we try to make each tag's transmission valid to avoid energy waste. So it is necessary for tag to distinguish the single tones efficiently in the uplink. When tag is dealing with carriers with different frequencies, PLM cannot provide a high-throughput link. Thus we need to enable tag with an efficient downlink for frequency identification. Our following work is motivated by the above limitations in BLE backscatter systems.

## III. EffBlue Design

### A. Overview

As shown in Fig. 1, we introduce a low-cost edge server that is built by commodity radios to provide excitation signals. The edge utilizes BLE periodic advertisement as carriers that possess higher carrier rates. Firstly, the tag identifies single tones among carrier packets to do time synchronization. However, the BLE carrier signal has diverse frequencies. Therefore, EffBlue introduces an efficient WiFi-ASK downlink to convey frequency information to tag. Our two-way communication supports tag to achieve efficient communication with commodity radios.

### B. Identify Excitation for Time Synchronization

*1) Why tag needs excitation synchronization:* RBLE proposes a direct frequency shift scheme, which improves reliability and supports direct communication with commodity BLE devices. In RBLE, single tones are shifted to two standard frequencies, representing different BLE symbols. When modulating on non-single tones, the backscattered symbol is uncontrollable. Thus commodity receiver cannot decode the backscattered signals. RBLE's modulation process is triggered by energy level, which could lead to invalid backscatter. As shown in Fig. 4, the BLE carrier signals are diverse, which includes target single-tone signal and non-single-tones. RBLE considers them as carriers from a single channel. So P1, P2, P5, and P6 are invalid due to wrong carriers. P4 is invalid due to unexpected carrier frequency. To improve the uplink modulation efficiency, the tag should time synchronize with the target single tones.

*2) Identify target excitation using packet length:* BLE band consists of 3 advertising channels and 37 data channels. Traditional advertisement signals (BLE 4) are on three advertising channels with random delays. In BLE 5, extended periodic advertising has a longer packet length and strict periodicity, which are better for backscatter. For an extended periodic advertisement application, BLE devices first advertise on primary channels (advertising channels in BLE 4), then transmit auxiliary packets on secondary channels (data channels in BLE 4). EffBlue edge generates periodic packets as excitations, which are denoted to blue color in Fig. 5. In the following, we are aiming to identify the single tones without interference from other excitation sources.

During advertising, *AUX_SYNC_IND* packets have fixed length and their time interval are strictly the same, which is denoted as the periodic interval as shown in Fig. 5. We could send these features to tag (downlink design is presented later in III-C), so it could count the time of high energy level and consider the packet as an extended periodic packet when the counter value equals $T_{pack}$. However, only when the packet ends can its length be determined. When tag determines that it is the expected carrier, it cannot modulate data anymore. So tag utilizes the periodicity of extended advertising to predict the next coming packet. As shown in Fig. 6, light blue boxes denote the extended periodic single tones. Counting from the start of packet, when current packet length equals $T_{pack}$, tag predicts that at $T_{interval}, 2T_{interval}, ..., nT_{interval}$ will the periodic packets come. This design leads to a problem that using one packet as an anchor will result in a time drift due to hardware imperfection. The periodic interval is specified in a


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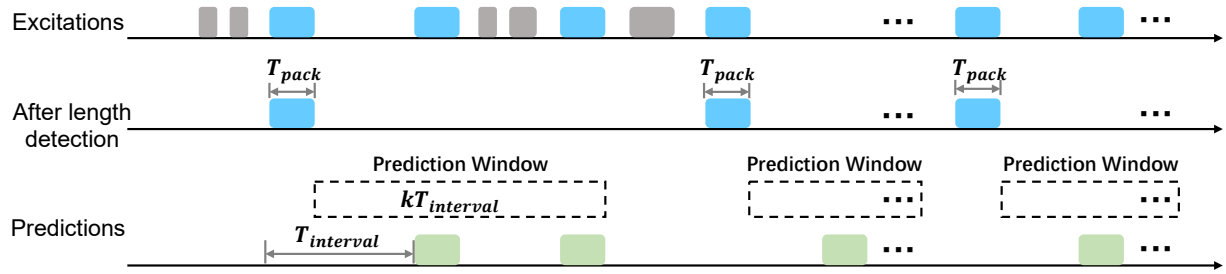
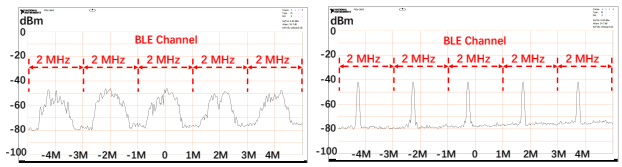



Fig. 9. Iterate-Sync. The key idea is to iterative predict the future single-tone carriers within a prediction time window.




(a) Non-single tones. (b) Single tones.

Fig. 10. The spectrum of non-single tone vs single tone of extended periodic advertisement packet in several secondary channels.

unit of 1.25ms. The actual time interval between two packets has microseconds of deviation compared to the theoretical value. The prediction of single tones will drift from the actual packet larger and larger. Fig. 8 shows the drifts of the predictions when sampled at every 5s.

*3) Prediction window:* We introduce Iterate-Sync to solve this issue. The key idea is to predict the future single-tone carriers iteratively within a limited time window. In the first stage, the tag counts the length of coming packets $T_p$. If $|T_p - T_{pack}| \leq e$, tag considers it as target periodic packets as shown in Fig. 9. Here $e$ denotes the error bound, which is used for toleration because clock accuracy or energy interference could affect the detected length. Using packet length can filter out most non-periodic packets. In the second stage, tag uses these valid packets to predict future periodic packets within a prediction window. And it can predict that future extended periodic packets are arriving at $T_{interval}, 2T_{interval}, ..., kT_{interval}$, where $k$ depends on the prediction window size. This prediction window size ensures that the accumulated drifts within this window are tolerated by the backscatter system. When the prediction window ends, tag will again use packet length to identify target excitation packets and start new prediction windows. Since the key of this prediction process is recording the time of packets and interval, this algorithm can be implemented using clocks and register resources with a space complexity O(n). When using a 5 MHz clcok, 32-bit registers are enough, which are easy to obtain in state-of-the-art platforms.

**Extension to BLE data carriers.** Our system can also be extended to utilize data packets as carriers. Periodic advertisement, which is our single-tone excitation, broadcasts extended data on secondary advertisement channels, which are traditional data channels. BLE data packets share the same hopping algorithm and channels with periodic advertisement, and the data carriers are periodic too. To support utilizing data packets, the BLE module in the edge can be programmed to be a sniffer, which acquires the hopping parameters of data carrier packets. And the remaining operations are the same and the edge transmits these parameters to tag to calculate carrier frequency.

### C. WiFi-ASK Downlink for Frequency Synchronization

In the above, with time synchronization, tag knows when single tones come. But the frequency of single tones is diverse. Fig. 10(a) shows the signal spectrum of *AUX_SYNC_IND* packets of extended advertising within a period. They are hopping on the secondary channels so that we can see several peaks in the band. To generate periodic single tones, we acquire the channel index of each extended advertisement packet and dewhiten the payload. Fig. 10(b) shows the results of dewhitening, which are diverse single tones. So tag should perform frequency synchronization before modulation. To enable excitation synchronization, the packet length, period, and frequency information are required. Thus tag needs an efficient downlink design. In each excitation interval, the tag decodes the downlink data to get the excitation frequency and then modulates the incoming excitation packet within this interval. The tag regenerates BLE packet on another static or dynamic target channel, which requires a detailed MAC layer design between tag and receiver. The key is the efficient downlink. For periodic advertisement excitations, the minimum hopping interval is 7.5 ms. The downlink data includes length (2B), interval (2B), channel map (5B), accessAddress (4B), eventCounter (2B), and hopIncrement (1B), which are parameters of hopping algorithm. The downlink throughput should be larger than 17 kbps.

Since the backscatter tag cannot directly decode frequency-modulated signals, state-of-the-art backscatter systems [10], [23], [24] usually utilize the envelope in the downlink. To generate downlink ASK messages, traditional methods adopt packet length modulation, which is of low efficiency. Interscatter [18] proposes a way that uses WiFi radios to generate amplitude-modulated signals. WiFi devices are commonly used in daily life. Constructing edge using a WiFi module will greatly reduce the cost. However, Interscatter only proposes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| Preamble (8 symbols) | Length (8 symbols) | Payload (0-256 symbols) | CRC (8 symbols) |
|---|---|---|---|

Fig. 11. WiFi-ASK downlink packet format.

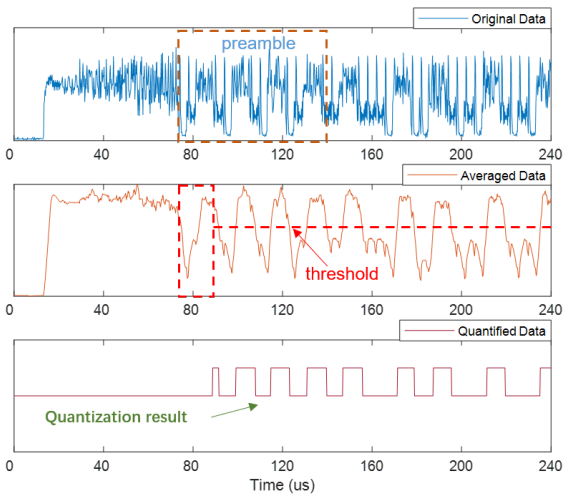


Fig. 12. WiFi-ASK downlink decoding procedure. The top figure shows the original envelope of the amplitude-modulated WiFi packet and it contains a downlink preamble in the middle of the packet. The middle figure shows the averaged data and the start of the preamble is used to determine the threshold to quantize the WiFi signal. The quantized results are shown in the bottom figure.

how to encode bits 0 and 1 with different WiFi symbols. To make this scheme work, we propose a complete design including synchronization, decoding, etc.

*1) Tx of WiFi-ASK signal:* The core idea of Interscatter is to transform 802.11g WiFi packets into amplitude-modulated signals by controlling the payload. OFDM symbols are classified into constant and random symbols. For random symbols, the energy is spread over the time domain. While in constant symbols, the energy is in the front part and the remains are zero. To generate constant symbols, we need to control the payload to make data bits constant before modulation. Reviewing the transmission process of WiFi packets, data bits will experience scrambling, encoding, and interleaving before modulation. If we generate constant data bits after scrambling, the final data will be constant too. Scrambling is an XOR operation between data bits and a particular scrambling sequence. This sequence is generated using a scrambling seed, which can be monitored using USRPs. With scrambling seed, we can set data bits to a particular value, making it constant after scrambling. We control the payload to generate constant and random symbols and get a WiFi signal with a different amplitude. The edge server encodes downlink data in a WiFi packet using ASK. The format of the downlink packet is shown in Fig. 11. The preamble field is used to identify the packet. The Length field indicates the length and the CRC field is used to check the correctness of transmission.

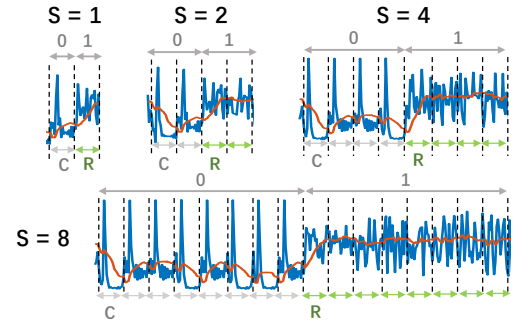


Fig. 13. WiFi-ASK Downlink symbol encoding scheme. We use $S = 1, 2, 4, 8$ WiFi symbols to encode bit '0' and bit '1'.

*2) Rx on tag:* We designed an RC filter circuit to extract the envelope. The quality of the envelope is related to the discharging speed of the capacitor. To extract the accurate envelope, the main parameter $\tau$ needs to satisfy: $1/f_c \ll \tau \ll 1/f_b$, where $f_c$ is the carrier wave's frequency, which equals to 2.4 GHz, and $f_b$ is the baseband frequency, which equals to 250 kHz. We use an ADC to sample the envelope. In 802.11g, each OFDM symbol takes $4\mu s$. The EffBlue tag samples the WiFi-ASK signal at a sampling rate of 5 Msps, thus there are 20 samples for each symbol, which would be enough for decoding the amplitude information. Fig. 12 shows the sampling result of envelope. The random symbols and constant symbols do not perfectly fit an amplitude-modulated signal. There are apparent fluctuations of amplitude within symbols because we are using WiFi radios instead of dedicated devices. We simply use an averaging window $L_a$ to filter the high-frequency component. The middle figure in Fig. 12 shows the envelope of WiFi signals after smoothing.

We can see that at the start of the WiFi signal, there is no apparent amplitude change compared to the remaining part. This is because the WiFi packet has a preamble, header, addresses, etc. Symbols containing the above part cannot be turned into constant symbols. So the beginning part is of constant amplitude. The downlink data is encoded in two amplitudes. We quantize the amplitude into binary form and then decide the modulated data in each downlink symbol window. Amplitude modulation is vulnerable to distance. The amplitude is related to the signal power in our rectifier. If the threshold is fixed, the amplitude of the whole packet could be larger or smaller than the threshold. The threshold should be adaptive. Thus we adopt a dynamic threshold. We transmit a known sequence of bits, like preamble in Bluetooth and WiFi, to do threshold control and symbol synchronization.

As shown in Fig. 12, we use two constant symbols to encode bit 0 and two random symbols to encode bit 1. Thus each downlink symbol lasts 8 $\mu s$. The Preamble is defined as an 8-bits sequence, 01010101. The Preamble field starts from the middle part of the WiFi signal as Fig. 12 shows. We use a 16 $\mu s$ window, named threshold window, to calculate the average of the first two downlink symbols in the Preamble. Since this window contains one random symbol and one constant


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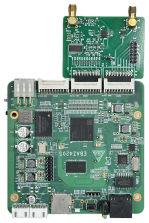

Fig. 14. Tag photo.

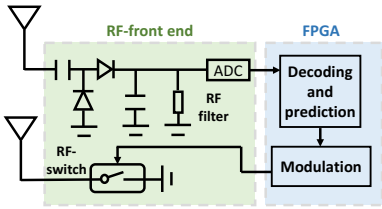


Fig. 15. Overview of tag circuit.

symbol, the average can be used as a threshold to quantize the remaining symbols. The quantized results are shown in the bottom figure in Fig. 12. After the Preamble, we group every 40 samples (two OFDM symbols with 20 samples each at 5 Msps) as a downlink symbol. Then we calculate the sum of these samples $D_{sum}$ within one symbol. Since the start point could drift slightly from the ground truth, there could be both 0 and 1 within one symbol. We decode symbol as bit 0 if $D_{sum} < 20$ and decode symbol as bit 1 if $D_{sum} > 20$. And during demodulation, if the preamble is not detected, demodulation will terminate. And we also add a CRC field at the end of the downlink packet to ensure that there is no bit error.

In the above, each ASK symbol consists of two WiFi symbols. We can also adopt redundant encoding to improve communication quality. Besides 2 WiFi symbols per downlink symbol, Fig. 13 also shows two different encoding coefficients: 4, 8, which corresponds to 16 $\mu s$, 32 $\mu s$ encoding rates. The edge server can notify the tag to adapt to different encoding rates.

**Extension to receiver-to-tag link.** The traditional two-way communication utilizes a bidirectional link between transmitter and receiver. To enhance the efficiency of two-way communication, our EffBlue tag can also be upgraded to support receiver-to-tag link. Currently, our edge can transmit messages to EffBlue tag. It is easy to modify or add a BLE module to the edge to sniff the BLE packets from the receiver and then the edge forwards BLE data to tag through the WiFi-ASK downlink. Thus this architecture enables the acknowledgment and control of data transmission between the tag and receiver.

## IV. Implementation

The off-the-shelf EffBlue prototype consists of two parts, a tag, and an edge server. The implementation is detailed as follows.

**Tag prototype.** The EffBlue tag consists of an RF front-end and an FPGA as shown in Fig. 14. The front-end consists of an RF filter as shown in Fig. 15, which is used to extract the envelope of the WiFi-ASK packet. The AD9235 ADC is used to sample the envelope and fed to FPGA for further processing. The processing logic is implemented on a ZYNQ XCZ7010 FPGA and it controls an ADG902 RF switch to backscatter. The power consumption of this FPGA-based prototype is 177.83 $mW$ in total. We also simulate an IC prototype using Cadence IC6.17 Virtuoso software. The estimated power consumption is 841.41 $\mu W$ as shown in Tab. I, which is mainly owing to the oscillator and ADC. We believe the power consumption of them will be further reduced with advanced techniques in the future.

TABLE I
Power consumption of tag prototypes

| | Digital core | Oscillator | RF-switch | ADC | Total |
|---|---|---|---|---|---|
| FPGA-based | 16mW | 85mW | 1.83mW | 75mW | 177.83mW |
| IC | 267uW | 300uW | 2.41uW | 272uW | 841.41uW |

**Edge server.** The edge server contains an excitation generator, a downlink transmitter, and a control logic. The excitation generator is built on Nordic NRF52840, running a modified Zephyr Periodic Advertising Application. The downlink transmitter is built on TI CC3200. These two parts are connected to the control logic through UART. The control logic is implemented using a Lenovo laptop and it forwards the information of the excitation generator to the WiFi-ASK transmitter. Since edge is mainly built on off-the-shelf chips, it can achieve a low cost. The NRF52840 module costs about $6. The CC3200 module costs about $10. The controller is a laptop, which can be replaced with low-cost MCUs (e.g., STM32) in future work due to low computational load.

**Experiment setup.** Our experiments are conducted in an indoor room. For excitation synchronization, the transmit power of the excitation signal generator is tuned to 8 dBm and connected with a 17 dBm gain amplifier. We fix the position of EffBlue tag and move the edge server along a straight line, and evaluate the detection precision of the excitation signal. Then we compare it with energy-based identification in RBLE. We also measure the synchronization accuracy through an oscilloscope.

For WiFi-ASK downlink, we set transmit power of CC3200 to 20dBm. And we measure downlink BERs and throughputs under the different downlink distances and compare it with PLM adopted in FreeRider and RBLE. We implement PLM on CC1352P using two different length packets: 26-byte and 34-byte long. The transmit power of CC1352P is set to 20dBm, the same as EffBlue downlink module. The PLM message includes a preamble to identify the start of message.

## V. Evaluation

In this section, we first conduct the micro-benchmark experiments and then measure the end-to-end performance.

### A. Excitation Synchronization

*1) Impact of Error Bound:* EffBlue first identifies the single-tone packet through packet length. Tag sets an error bound for detection. We evaluate the impact of error-bound size.

We make edge generate packets with fixed lengths and measure the detection rate. The error bound size is set to 1 $\mu s$, 2 $\mu s$, 4 $\mu s$, 8 $\mu s$, 16 $\mu s$. We move the edge server along a straight line. In Fig. 16(a), when edge-to-tag distance is 10 cm, more than 98 percent of packets can be recognized by tag.


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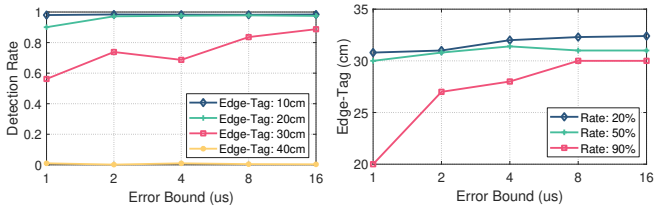
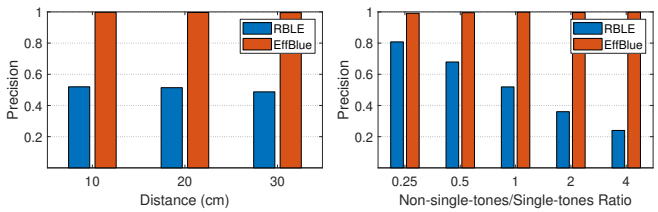


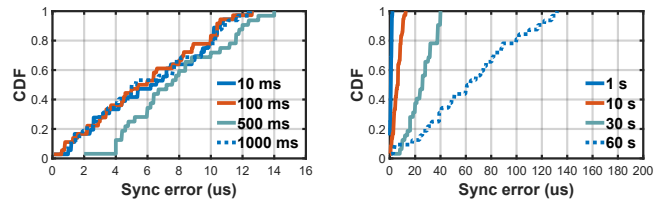

(a) Detection Rate Vs Error bound. (b) Edge-Tag distance Vs Error bound.

Fig. 16. Impact of the error bound size on packet detection based on length.



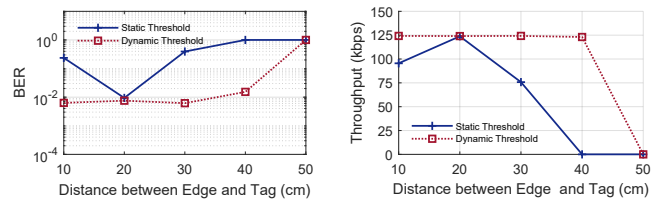

(a) Detection precision under different distances. (b) Detection precision under different Non-single tones/Single tones ratio.

Fig. 17. Impact of edge-tag distance and non-single tones ratio.




(a) Different excitation rates. (b) Different prediction window sizes.

Fig. 18. EffBlue excitation synchronization error.




(a) BER comparison in different edge-tag distances. (b) Throughput comparison in different edge-tag distances..

Fig. 19. Performance comparison of static quantization threshold and dynamic quantization threshold.

At 20 cm and 30 cm, when the error bound increases from 1 $\mu s$ to 2 $\mu s$, there is an improvement, because when the energy level is not strong, the detected length could fluctuate more than 1 $\mu s$. While at a 40 cm distance, less than 2 percent of packets can be detected. Fig. 16(b) shows the maximum edge-to-tag distance under different detection rates. When the error bound increases from 1 $\mu s$ to 2 $\mu s$, the maximum distance that enables 90% rate has a large improvement. When the error bound is larger than 2 $\mu s$, the improvement is limited. In the following experiments, we set the error bound to 8 $\mu s$.

*2) Detection Precision:* The prior energy-based identification method that RBLE adopts will be triggered by non-single tones, leading to false positive cases. We use Precision $= \mathrm{TP}/(\mathrm{TP}+\mathrm{FP})$ to evaluate the performance of EffBlue with RBLE under different scenarios.

**Impact of distance.** We set the ratio of non-single tones and single tones in the excitation signal to be 1:1. We move the edge server along a straight line and calculate the detection precision. In Fig. 17(a), RBLE achieves 51.9%, 51.4%, and 48.7% at 10, 20, and 30 cm respectively. While for EffBlue, it can keep a 99% detection precision under different distances. Its excitation identification mechanism greatly reduces the false-positive cases that RBLE suffers.

**Impact of non-single tones/single tones ratio.** The detection precision is mainly related to the ratio of non-single tones. We fix the edge-tag distance at 20 cm and change the ratio of non-single tones to see the performance. Fig. 17(b) shows that when the ratios are at 0.25, 0.5, 1, 2, 4, RBLE's detection precision is at 80.7%, 67.8%, 51.9%, 36%, 24% respectively. With the increasing proportion of non-single tones, RBLE's detection precision becomes smaller. This confirms that RBLE cannot distinguish the non-single tones. However, EffBlue keeps 99%+ detection precision because it will not be triggered by the non-single tones.

*3) Synchronization Error:* EffBlue uses detected single tones to predict future carriers for modulation. We measure the deviation between predicted single-tone packets and actual packets as synchronization errors under different excitation rates and prediction window sizes.

**Different excitation rates.** The prediction window size of EffBlue is set to 10s. Then we set the single-tone packet interval to 10 ms, 100 ms, 500 ms, and 1000 ms, and measure EffBlue's synchronization error. Fig. 18(a) shows that the distribution of 10, 100, and 500 ms cases are almost the same. And 1000 ms case has a slight difference from the others. However, all the synchronization errors are within 12 $\mu s$. This shows that excitation rates do not affect the synchronization error when the prediction window size is fixed because the accumulated drift only depends on time.

**Different prediction window sizes.** The predicted carrier packet drifts within the prediction window. So the window size affects the synchronization errors. We evaluate synchronization performance under different prediction window sizes of 1 s, 10 s, 30 s, and 60 s. The single-tone packet interval is set to 100 ms and the edge-to-tag distance is set to 20cm. We randomly sample the synchronization results. Fig. 18(b) shows that when window sizes are 1 s, 10 s, 30 s, 60 s, 96% errors are within 2 $\mu s$, 12 $\mu s$, 40 $\mu s$, and 128 $\mu s$ respectively. And 50% errors are within 0.9 $\mu s$, 7 $\mu s$, 26 $\mu s$, and 64 $\mu s$. With increasing window size, the synchronization errors become larger. With a window size of 10s, 99% errors are within 12$\mu s$, which is 1 to 2 bytes long in a BLE packet. This deviation can be tolerated in EffBlue. So we take a 10s length window size in the remaining evaluations.



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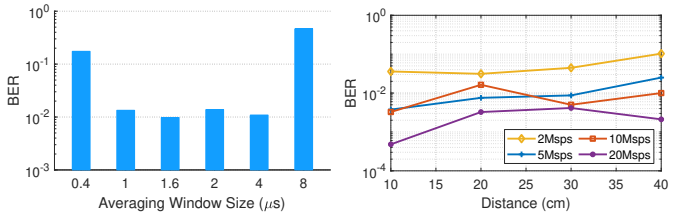
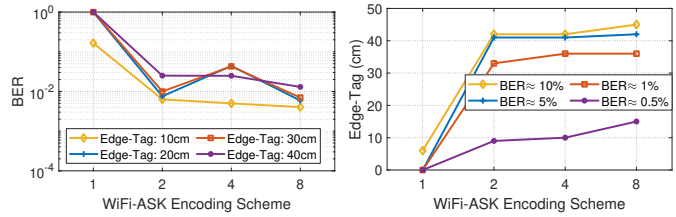



(a) BER comparison under different averaging window size.

(b) Throughput comparison in different edge-tag distances.

Fig. 20. Impact of averaging window size and sample rate.




(a) BER Vs Encoding scheme.

(b) Edge-Tag distance Vs Encoding scheme.

Fig. 21. Impact of WiFi-ASK encoding scheme on BER and distances.

### B. WiFi-ASK Downlink

We evaluate EffBlue's downlink performance from the following aspects: quantization threshold, averaging window length, sample rates, and WiFi-ASK symbol length.

**Quantization threshold.** The static threshold takes the value that performs well at 20cm. The dynamic threshold is calculated through the preamble. We move edge along a straight line and measure BER and throughput at different edge-tag distances. Moreover, we set ASK symbol length to 8 $\mu s$, sample rate to 5 Msps, and averaging window length to 1.6 $\mu s$. As shown in Fig. 19(a) and 19(b), the static threshold case achieves 1% at 20 cm. However, when tag moves to 10 cm, 30 cm, and 40 cm, the BERs are larger than 10%, which leads to a sharp drop of throughput in static threshold cases. Because the static threshold is not able to correctly quantify the amplitude. In dynamic threshold cases, the BERs keep at around 1%, and their throughput stays at around 125 kbps. Compared to the static threshold, the dynamic threshold can adapt to different strengths of downlink WiFi-ASK signal and it performs better at different communication distances.

**Averaging window size.** The extracted WiFi-ASK signals of envelope detector include high-frequency components which affect the decoding performance. So EffBlue tag adopts an averaging window to smooth the envelope. This window size would impact the decoding performance. Thus we evaluate the downlink BER under different averaging window sizes at 5 Msps sample rate, S = 2 encoding scheme. As shown in Fig. 20(a), the averaging window size is set from 0.4 $\mu s$ to 8 $\mu s$. At 0.4 $\mu s$ and 8 $\mu s$, the downlink BERs are larger than 10%, which is not acceptable. When the size is too small, the high-frequency signal remains. While with large size, the ASK information could be removed. When window size takes 1 $\mu s$, 1.6 $\mu s$, 2 $\mu s$, 4 $\mu s$, the downlink BERs are all around 1%. In the remaining experiments, we set the window size to 1.6 $\mu s$.

**Sample rate.** The ADC sample rates determine the resolution of extracted WiFi-ASK signals. We evaluate the BERs of different sample rates from 2 to 20 Msps at different edge-to-tag distances. The downlink encoding scheme is set to S = 2. Fig. 20(b) shows an overall trend that BERs are increasing with distance. Among all cases, 2 Msps perform worst, whose BERs reach 3% at 10cm+ distance. A small sample rate leads to fewer samples in each WiFi-ASK symbol, which is vulnerable to interference. At 40cm distance, its BER achieves at around 10%, almost 100x than other sample rates. Sample rates of 5 Msps, 10 Msps, and 20 Msps perform better. They have an average BER of less than 1% within 30cm. When sample rates increase from 2 Msps to 5 Msps, we could observe that BERs are around 10x smaller. When sample rates increase from 5 Msps to 10 Msps, the improvement is not as large as 2 to 5 Msps. The sample rate of 20 Msps performs best among all distance cases. At 10cm, its BER can be as low as 4‰. 20 Msps is a rather high rate that will consume more resources and power and 2 Msps is too bad to adopt. Thus 5 Msps would be a proper choice for common use.

**Encoding scheme.** To see the impact of different encoding schemes of WiFi-ASK symbols on downlink BER and distance, we conduct two sets of experiments. In Fig. 21(a), we can observe that with WiFi-ASK symbol increasing from S = 1 to S = 2, BERs can be greatly reduced. At S = 1, the downlink almost cannot work. When S = 2, BERs under different distances are around 1%. When the encoding scheme increases from S = 2, BERs change little, except at S = 4, which has an increase and then follows a drop. And S = 8 does not improve much compared to S = 2. Fig. 13(b) also shows that increasing S = 1 to 2 could benefit a lot. But after S = 2, the gain on distance is small and the dominant factors that affect BERs are not the encoding scheme, but others like sample rate.

**WiFi-ASK vs PLM.** To evaluate the end-to-end performance, we measure downlink BERs and throughput and compare it with PLM scheme. Here, we calculate the throughput based on the symbol rate. We set the sample rate to 5 Msps, and encoding scheme S to 2. The distance between edge server and tag increases from 5 cm.

Fig. 22(a) shows that BERs are increasing as distance increases. Within 30cm, EffBlue's BERs are less than 1%. When the distance is larger than 45cm, both WiFi-ASK and PLM cannot receive downlink packets anymore. PLM performs slightly better than WiFi-ASK within 35cm. However, the BER of PLM has a larger jittery than EffBlue. Because when one bit of PLM is missed or inserted, the remains will also be decoded incorrectly since the order of the bits is corrupted. So a small decoding mistake could cause large errors. However, improving the encoding scheme of PLM has beyond this paper. Fig. 22(b) shows that EffBlue has a throughput of around 125 kbps within 40cm and PLM only achieves 4 kbps. Because each PLM symbol takes about 240 $\mu s$. The length of WiFi-



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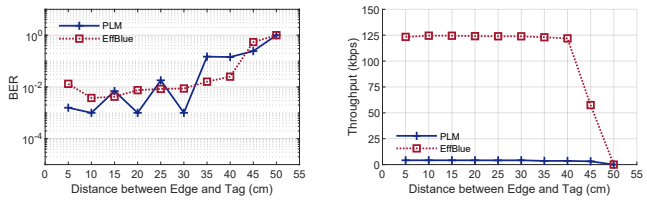
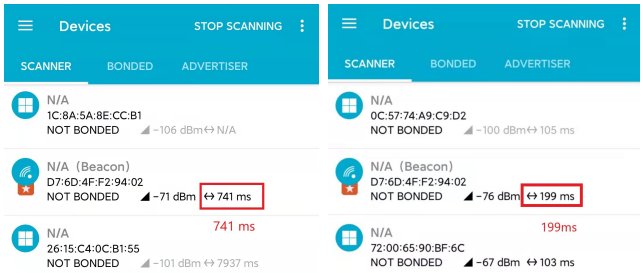



(a) BER. (b) Throughput.

Fig. 22. WiFi-ASK downlink BER and throughput across different edge-to-tag distance.




(a) RBLE. (b) EffBlue.

Fig. 23. Scanning results of Tag's Beacon packets on nRF Connect App.

ASK symbol is 8 $\mu s$, which is much smaller than PLM. So EffBlue has a 25x throughput gain over PLM.

Our edge achieves a downlink distance that is less than the typical reading range of RFID. This distance gap is mainly due to three reasons. First, EffBlue tag extracts the high-frequency signal by the output voltage samples of the RC filter, which has lower SNRs. Second, the WiFi-ASK signals are at 2.4 GHz while the RFID is operating at 900 MHz. Our downlink signal has shorter wavelengths, which brings less energy, resulting in low output voltage at the detector. Third, our edge uses a 3 dBi omni-directional antenna at the WiFi modules, which is less than the 9 dBic of patched antennas on a typical RFID reader. To enhance the edge downlink distance, we can increase the transmission power and use antennas with larger gain.

**Energy Consumption.** In EffBlue design, the ADC is used to sample the WiFi-ASK signal periodically. We have evaluated and optimized the sample rate to reduce computing complexity. Tag performs sampling periodically due to the synchronization drift. We can build a model for the drift and compensate the synchronization offset on tag to avoid frequent sampling. For future IC design, several recent work have proposed advanced ADCs [25]–[28] that can achieve a sampling rate of tens of Msps with a consumption of only tens of $\mu W$, which would greatly improve the whole consumption.

### C. Application

Based on EffBlue's efficient two-way design, we introduce a Beacon application. We set the interval of periodic single tones to 200 ms and let tag identify and modulate on single tones to generate BLE Beacons. We use the nRF Connect App on a MI 11 Android phone to scan tag's signal. Fig. 23(a) shows that the scanned advertising interval of RBLE Beacon is 741 ms. RBLE tag will modulate each coming packet. Modulation on non-single tones would be invalid. Meanwhile, RBLE cannot sense the carrier frequency. So modulation on unexpected frequencies would be invalid too. Thus the scanned interval could be up to 741 ms. For EffBlue, the scanned interval of Beacon is 199 ms as shown in Fig. 23(b). Because tag can identify each single-tone packet and its carrier frequency, making each modulation valid. So EffBlue tag can transmit valid Beacons every 200 ms.

## VI. Related Work

In the past decades, backscatter communication has attracted much interest and great efforts have been made to turn backscatter into general-purpose communication in IoT networks. Ambient backscatter [1] is the first system that adopts ambient TV signals as carriers. Later, lots of backscatter systems that adopt ambient signals as carriers emerge, such as WiFi [6], [12], [13], [29]–[34], BLE [17], [21], [35], ZigBee [17], [36], LoRa [37], [38], LTE [39], [40] FM [2], light, etc. Compatibility with commodity radios reduces the deployment cost compared to traditional dedicated devices. Beyond this, backscatter communication also enables a wide range of passive applications [3], [7], [11], [41]–[48].

Among backscatter systems, BLE backscatter has experienced a huge development. Ble-backscatter [16] backscatter new BLE packets on the continuous wave. Later FreeRider [17] extends the codeword translation to BLE, which adopts BLE packets as excitation instead of a continuous wave. And it also introduces a PLM downlink, which is implemented on commodity devices. Then RBLE [21] improves bit error rate and throughput of uplink. RBLE's modulation is triggered by the energy level. Meanwhile, it adopts a PLM downlink too. Along this research line, EffBlue is aiming to address the problem of inefficiency downlink and false excitations in the uplink.

## VII. Conclusion

We present EffBlue, an efficient two-way backscatter system that adopts an edge server to help enable efficient downlink and uplink of tag. We introduce an accurate synchronization scheme to efficiently eliminate non-single tones among periodic carriers. We also design a new symbol-level WiFi-ASK downlink which can be decoded by a simple envelope detector. Comprehensive experiments demonstrate that EffBlue achieves high single-tone identification precision and significant downlink throughput gains. EffBlue provides an efficient two-way design for general-purpose backscatter systems, thus opening the door to further interaction with commodity devices.

## VIII. Acknowledgment

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